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LETTER TO THE EDITOR

A purely L^2 method for calculating resonance widths†

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Abstract. We propose a new method for calculating the widths of atomic and molecular resonances. The method uses only square-integrable functions to approximate the non-resonant scattering solutions and employs Stieltjes-moment-theory techniques to extract a continuous approximation for $\Gamma(E)$ from a discrete representation of the background continuum. The method has been applied to several well known resonances in He, He⁻ and Mg⁻ with encouraging results. The proposed method is easily applied to molecular problems.

During the past few years, several techniques have been proposed for calculating the widths of autoionising and autodetaching resonance states of atoms (Drake and Dalgarno 1971, Temkin *et al* 1972, Bhatia *et al* 1973, Nicolaides 1972, Hazi and Fels 1971, Bain *et al* 1974, Hickman *et al* 1976, Oppenheimer and Doyle 1976). In spite of these theoretical efforts the calculation of resonance widths for molecules is far from routine, and only a few *ab initio* calculations have been reported in the literature (Miller *et al* 1972, Hickman *et al* 1977, Cohen and Lane 1977, Docken unpublished). In the case of molecules, the difficulty stems from the non-spherical nature of the potential which describes the electron-target interaction. For example, many of the methods utilise the 'golden-rule' formula for the width (Feshbach 1962)

$$\Gamma(E) = 2\pi |\langle \phi_r | H - E | \Psi_E^+ \rangle|^2 \quad (1)$$

where ϕ_r is a square-integrable (L^2) function representing the resonance, and Ψ_E^+ is an energy-normalised scattering function corresponding to the background continuum. Since a rigorous calculation of Ψ_E^+ for even a diatomic molecule (or ion) is very difficult, previous *ab initio* calculations for molecules utilised undistorted Coulomb waves. On the other hand, molecular bound-state or resonance wavefunctions can be obtained more easily, even when one includes many-electron correlation effects in the calculations. Thus, it would be highly desirable and practical to have a purely L^2 method which required only molecular-structure codes for calculating the width of molecular resonances.

In this letter, we propose such a method which employs only L^2 functions to approximate the non-resonant scattering function Ψ_E^+ , and which uses Stieltjes-moment-theory techniques‡ to extract the width, $\Gamma(E_r)$, from a discrete representation of the background continuum. Since the method avoids the explicit construction

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‡ An extensive review of this method and a compilation of appropriate references is given by Langhoff (1974).

of continuum wavefunctions, many-electron effects can be easily incorporated into the wavefunctions via standard configuration-interaction (CI) techniques. The proposed method has been tested on several well known Feshbach-type resonances in He and He^- and the low-energy shape resonance in Mg^- with very encouraging results.

The first step of the calculation involves finding a suitable approximation for the resonance state. The function ϕ_r is obtained by diagonalising the full many-electron Hamiltonian in a basis set of L^2 functions from which configurations representing the decay channel (target + ionised electron) have been projected out, i.e. ϕ_r satisfies

$$(Q_0 H Q_0 - \epsilon_r) \phi_r = 0. \quad (2)$$

For many-electron targets, the projection operator P_0 ($P_0 = 1 - Q_0$) projects onto only an approximate representation of the decay channel, since the exact target states are not known. However, using reasonable approximations for the target states, such as SCF wavefunctions, ensures that the desired resonance state ϕ_r is the lowest solution of (2).

Next, two new projection operators, Q and P are defined according to

$$Q = \sum_i |\phi_{ri}\rangle \langle \phi_{ri}| \quad (3)$$

and

$$P = 1 - Q \quad (4)$$

where the index i runs over all the resonances of a given symmetry under consideration†. Thus, P space contains not only those configurations which represent the decay channel (originally in space P_0) but also the *higher* non-resonant solutions of $Q_0 H Q_0$ which are associated with higher excited states of the target. We have found that this repartitioning of the spaces is essential in order to incorporate, as fully as possible, many-electron correlation and channel-coupling effects into the non-resonant continuum wavefunction Ψ_E^+ . In the case of many-electron targets (e.g. He^- , Mg^-), such effects must be accounted for if accurate widths are desired.

In order to obtain an approximation for the non-resonant continuum, the operator PHP is diagonalised in the basis of *all* L^2 configurations. The resulting eigenfunctions‡, which are orthogonal to $\{\phi_{ri}\}$, satisfy

$$\langle \chi_n | PHP | \chi_m \rangle = \delta_{nm} \epsilon_n \quad (5)$$

and

$$\langle \chi_n | \chi_m \rangle = \delta_{nm}. \quad (6)$$

The set $\{\chi_n, \epsilon_n\}$ forms a discrete representation of the background continuum, and near the nuclei, each χ_n approximates Ψ_E^+ at $E = \epsilon_n$ (apart from an overall normalisation factor). Since the matrix element in (1) involves a bound wavefunction ϕ_r localised near the nuclei, it has no significant contribution from the asymptotic region where

† In the case of neutral systems, one may wish to consider simultaneously several members of a Rydberg series of autoionising states which converge to an excited state of the ion, e.g., (2s, 2p), (2s, 3p), (2p, 3s), ... 1p in He.

‡ Of course, the resonance functions are also eigenfunctions of PHP , by construction, but with zero eigenvalues.

$\{\chi_n\}$ fail to approximate Ψ_E^+ (Miller 1970). Consequently, one may use $\{\chi_n\}$ to calculate the 'width matrix elements':

$$\gamma_n = 2\pi |\langle \phi_r | H | \chi_n \rangle|^2 \quad n = 1, 2, \dots \quad (7)$$

However, γ_n cannot be used directly to approximate $\Gamma(E = \epsilon_n)$ in (1), because χ_n is unit normalised (see (6)), whereas Ψ_E^+ is energy normalised.

Instead, we use Stieltjes moment theory (Langhoff 1974) to extract the appropriately normalised continuum information from the pseudospectrum $\{\gamma_n, \epsilon_n\}$. Let us consider the cumulative function $F(E)$, which is defined as

$$F(E) = \int^E dE' \Gamma(E') \quad (8a)$$

or

$$\Gamma(E) = \frac{dF}{dE}. \quad (8b)$$

The $\{\gamma_n, \epsilon_n\}$ can be used to furnish a histogram approximation to $F(E)$ in the form

$$\tilde{F}(E) = 0 \quad 0 \leq E \leq \epsilon_1 \quad (9a)$$

$$\tilde{F}(E) = \sum_{n=1}^k \gamma_n \quad \epsilon_k < E < \epsilon_{k+1}. \quad (9b)$$

At the rise points, $E = \epsilon_k$, $F(E)$ is approximated by

$$\tilde{F} = \frac{1}{2} [\tilde{F}(\epsilon_k +) + \tilde{F}(\epsilon_k -)] = \sum_{n=1}^{k-1} \gamma_n + \frac{1}{2} \gamma_k. \quad (10)$$

The development in (9) is analogous to evaluating the integral in (8a) by numerical quadrature, where the eigenvalues $\{\epsilon_n\}$ are the points associated with an equivalent quadrature (Heller *et al* 1973). The Stieltjes derivative at the half-way points $E_n = \frac{1}{2}(\epsilon_n + \epsilon_{n+1})$ is given by

$$\tilde{\Gamma}(E_n) = \frac{\gamma_n + \gamma_{n+1}}{2(\epsilon_{n+1} - \epsilon_n)} \quad n = 1, 2, \dots \quad (11)$$

The result in (11) provides a histogram approximation to the width function $\Gamma(E)$.

In practice, one does not work directly with the pseudospectrum

$$\{\gamma_n, \epsilon_n, n = 1, \dots, N\}.$$

Instead, one first performs a moment analysis to obtain a 'smoothed' spectrum

$$\{\tilde{\gamma}_n^{(M)}, \tilde{\epsilon}_n^{(M)}, n = 1, \dots, M\}$$

whose elements are uniquely determined by the first $2M$ inverse-power moments of the pseudospectrum, provided $M \leq N$. The calculations are repeated for different values of M ($M \ll N$) until mutually consistent results are obtained. The details of the moment analysis and the algorithm required for obtaining the 'smoothed' spectrum are given elsewhere (Langhoff and Corcoran 1974). The reader should note that the Stieltjes derivative in (11) provides an approximation to $\Gamma(E)$ only at discrete energies. To obtain the physical width, $\Gamma(E_r)$, an interpolation of $\tilde{\Gamma}(E_n)$ would be

required. However, previous application of the Stieltjes imaging procedure to the calculation of photoionisation cross sections (Hazi and Rescigno 1977) has shown that an accurate and continuous approximation to dF/dE may be obtained by fitting the approximate cumulative function in (10) to an appropriate analytic function of E , which can be differentiated directly at any energy (see (8b)). In the present work, we used both the Stieltjes derivative and analytic fits of $\tilde{F}(E)$ to calculate the resonance widths.

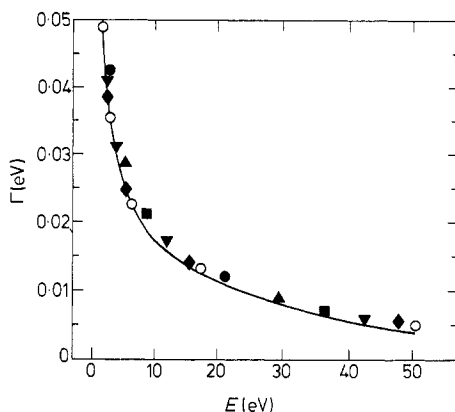


Figure 1. The width as a function of energy for $(1s2s^2)^2S$ state of He^- . Full curve: $\Gamma(E)$ from analytic fit of cumulative function. Individual points are Stieltjes derivatives from the first M moments. ●, $M = 3$; ▲, $M = 4$; ■, $M = 5$; ▼, $M = 6$; ◆, $M = 7$; ○, $M = 8, 9$.

As an illustration, figure 1 shows the width function calculated† with these two procedures for the $(1s2s^2)^2S$ resonance in He^- . The Stieltjes values obtained with various number of moments ($2M$) fall on a reasonably smooth curve and are mutually consistent with the continuous approximation for $\Gamma(E)$.

In addition to the $(1s2s^2)^2S$ resonance of He^- , we have applied the method to several well known autoionising states of He, and the low-energy $(3s^23p)^2P$ shape resonance in Mg^- . In all these cases we employed moderately sized CI wavefunctions (with 154 to 251 configurations) to include electron-correlation effects for the active valence electrons. In the case of Mg^- , virtual excitations of only the 3s and 3p electrons were considered—the $1s^22s^22p^6$ core was described by an SCF wavefunction. Table 1 compares the computed resonance parameters to those obtained in previous calculations. In all cases, except for the very broad resonance in Mg^- , we report the ‘unshifted’ eigenvalue ϵ_r (see (2)) as the resonance energy, E_r . As table 1 shows the present method yields resonance widths which agree with the values obtained with more traditional procedures for both Feshbach-type and shape resonances. The method is also applicable to other than the lowest autoionising state of a given symmetry (e.g. the $(2s3p)_+^1P$ He state in table 1). We wish to emphasise, however,

† In this case, we used a basis of 13s, 4p and 3d orbitals from which we constructed a total of 251 configurations. A complete discussion of the partitioning of the basis set into P and Q space, and the details of the Stieltjes computations would be too long to present here and will be published elsewhere. However, the reader should note that it is essential to use a large number of L^2 functions to describe the scattered electron in order to obtain meaningful results.

Table 1. Comparison of computed resonance parameters.

Resonance	Present method		Previous calculations	
	E_r (eV)	Γ (meV)	E_r (eV)	Γ (meV)
He (2s2p) 3P	58.31	8.3	58.30 58.31	8.4 ^a 10 ^b
He (2s2p) 1P	60.19	36	60.13 60.13	38 ^c 37 ^b
He (2s3p) 1P	63.67	8.3	63.68	8.0 ^a
He ⁻ (1s2s ²) 2S	19.40	11.5	19.40 19.39	12.1 ^d 12.1 ^c
Mg ⁻ (3s ² 3p) 2P	—	200	0.15	230 ^f

^a Bhatia and Temkin (1969).^d Bain *et al* (1974).^b Drake and Dalgarno (1971).^e Junker and Huang (unpublished).^c Bhatia *et al* (1973).^f Robb (unpublished).

that the present formalism is not intended to provide highly precise widths for atomic resonances and thus it is not competitive with other methods which are based on rigorous calculations of the total or background scattering wavefunctions. The formalism is intended for molecular resonances, where other computational schemes become prohibitively difficult.

Recently, Hickman *et al* (1976) formulated a similar method for calculating resonance widths which is also based on the partitioning of a set of purely L^2 functions into resonant and non-resonant subsets. They used the density of states in the continuum (Hazi and Taylor 1970) to normalise correctly the L^2 approximations to Ψ_E^+ in (1). However, their method has the disadvantage that it requires the diagonalisation of PHP in several different basis sets in order to shift one of the background eigenvalues (ϵ_n in (5)) near the resonance energy. Such a procedure becomes prohibitively expensive when large CI wavefunctions are employed. In the present development, the Stieltjes-moment-theory technique provides both the required renormalisation of the approximate background solutions and the entire non-resonant continuum is a single diagonalisation of PHP . In a somewhat different context, Langhoff and co-workers (private communication) have also used the Stieltjes-imaging procedure to calculate the photoabsorption profiles of isolated atomic autoionising states in the continuum.

Finally, we wish to point out that the present formalism can be used to determine the resonance 'shift' which is given by

$$\Delta(E) = (2\pi)^{-1} \mathcal{P} \int \frac{\Gamma(E')}{E - E'} dE'. \quad (12)$$

Once an analytic approximation for $\Gamma(E)$ is available, (12) can be evaluated directly.

The application of the proposed method to molecular resonances is in progress, and the results will be reported in due course.

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